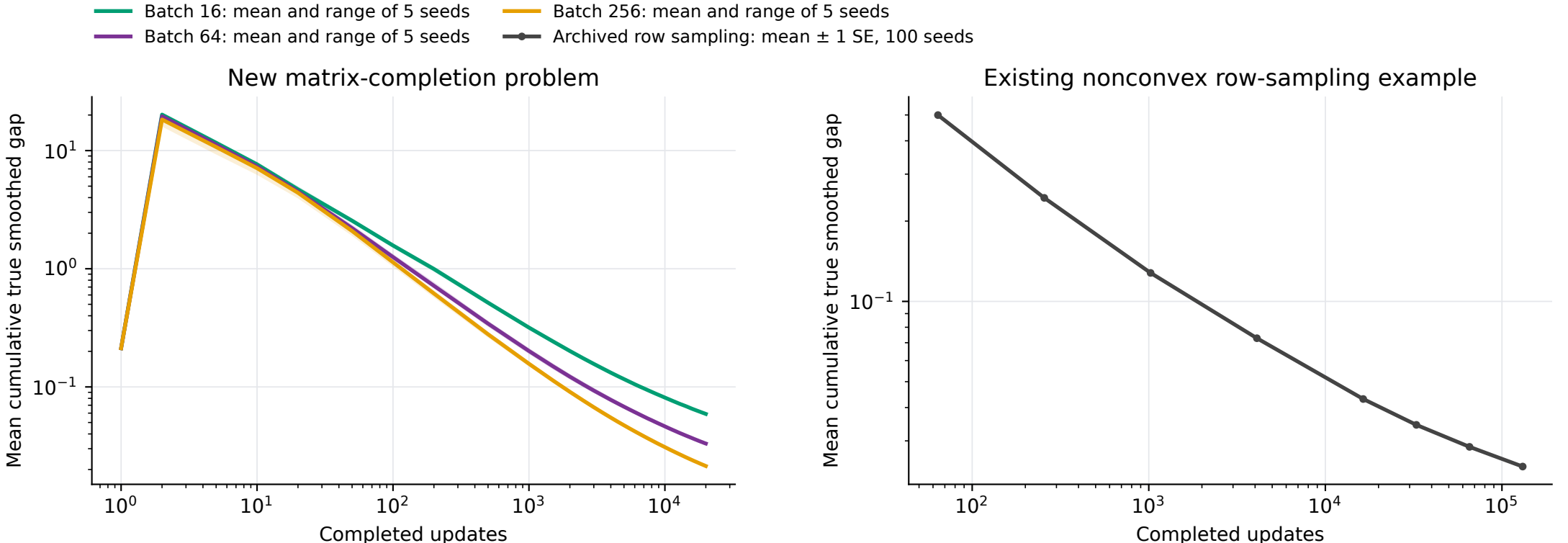
Theorem statistic: true averaged smoothed FW gap

Every pre-update iterate contributes its own exact gap. Expectations estimated across independent seeds



Left: SFRAMES with $\gamma_0=\rho_0=1$ and selected β_0; mean of per-run arithmetic averages over all $k=0,\ldots,K-1$.

Right: archived 1,024-variable construction, batch1. Its global optimum and full gradient are available analytically.

These are different problems and are not a solver ranking. Declining empirical gaps illustrate the theorem; they do not prove its asymptotic rate.